

Name: Devansh Singh
PRN: 25070122072
Batch: A-2
Divison: A

Subject: Programming with Java

ASSIGNMENT No : 14

TITLE : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Main1.java
Error : Cannot divide by zero.
Error : / by zero
Program continues after exception handling.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Main2.java
Error : NumberFormatException occurred.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Main3.java
Outer try block started
Inner try block started
Error + / by zero
Caught NullPointerException in outer try block : Cannot invoke "String.length()" because "<local1>" is null
Finally block of outer try executed
Program continues after nested try-catch blocks
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create an ATM program handling invalid withdrawal amount using try-catch.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp1.java
Enter withdrawal amount: 1000
Withdrawal Successful.
Remaining Balance: Rs.9000.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp1.java
Enter withdrawal amount: -1
Error: Invalid withdrawal amount!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp1.java
1Enter withdrawal amount: 100000
Error: Insufficient balance!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp2.java
Enter number of products to but : 3
Purchased!
Remaining Products : 7
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp2.java
Enter number of products to but : -1
Error: Invalid Product Amount.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp2.java
Enter number of products to but : 0
Error: Please enter amount more than 0.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp14/Exercise/Exp2.java
Enter number of products to but : 100
Error: Not enough products in stock to buy.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This experiment helped in understanding the concept of exception handling using `try`, `catch`, nested `try`, and multiple `catch` blocks in Java. It demonstrated how runtime errors can be detected and handled gracefully without causing abnormal program termination. Using appropriate exception handling techniques improves program reliability, enhances user experience, and makes applications more robust and maintainable. These concepts are widely used in banking systems, e-commerce applications, file handling, database operations, and other real-world Java applications where secure and efficient error management is essential.